

CSL2050 - Pattern Recognition and Machine Learning

Fruits Classification Project Report

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Abstract

This report represents a systematic and structured approach to fruit classification using traditional machine learning methodologies. Utilizing the well-curated Fruits-360 dataset, the project focuses on extracting discriminative visual features through Colour Histograms and Histogram of Oriented Gradients (HOG), capturing essential information related to color distribution, shape, and texture of fruit images.

Multiple classifiers like K-Nearest Neighbors (KNN), Decision Trees, Naive Bayes, Logistic Regression, and Support Vector Machines (SVM) were trained and evaluated on these characteristics to assess their effectiveness in varying representations. The study demonstrates that feature selection plays a pivotal role in classification performance, with color histograms offering superior accuracy for certain models, while HOG features provided complementary strengths in others.

Through extensive experimentation and comparative analysis, the report outlines classifier-specific behaviors, highlights challenges such as class imbalance and intraclass similarities, and documents preprocessing strategies employed to enhance model robustness. The findings offer valuable insights into the applicability of conventional machine learning techniques for image-based classification tasks in real-world scenarios.

Keywords: Fruit Classification, Machine Learning, Colour Histogram, HOG, SVM, KNN, Image Processing

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1 Introduction

Fruit classification plays a vital role in automating tasks in agricultural and retail industries. With the increasing availability of high-quality image datasets and the evolution of machine learning techniques, accurate recognition of fruit categories has become feasible. In this project, we focus on classifying fruits from the Fruits-360 dataset using traditional machine learning algorithms.

1.1 Objective

The objective of this project is to develop an efficient fruit classification system using traditional machine learning techniques. Leveraging the Fruits-360 dataset, the goal is to extract meaningful features such as color and texture from fruit images and evaluate multiple classifiers—including K-Nearest Neighbors, Decision Trees, Naïve Bayes, Logistic Regression, Support Vector Machines, for their performance in recognizing various fruit categories. The project also aims to analyze the effectiveness of different feature extraction methods, such as Color Histograms and Histogram of Oriented Gradients (HOG), in enhancing classification accuracy.

1.2 Dataset Overview

This dataset is a well-organized collection of high-resolution, labeled images across 141 fruit categories, totaling 94,110 images for both training and testing. Each fruit is captured from multiple angles, offering a wide range of visual perspectives to help models generalize better. The dataset also reflects natural variations in size, shape, and appearance, making it ideal for fine-grained image classification tasks. All images are preprocessed and standardized in size and format, ensuring consistency and ease of integration into machine learning workflows. With clear splits for training and testing, it supports reliable model evaluation and performance benchmarking. This dataset serves as a strong foundation for fruit recognition, classification, and related applications.

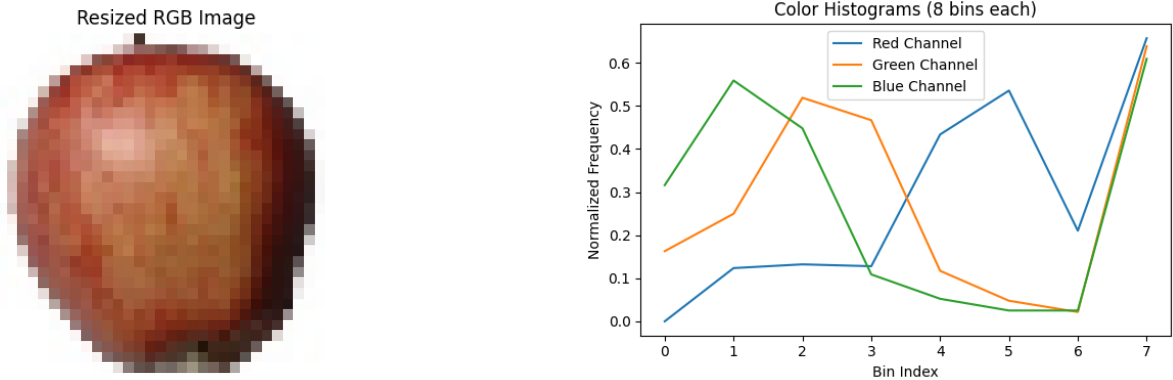
2 Methodologies Applied

2.1 Feature Extraction Techniques

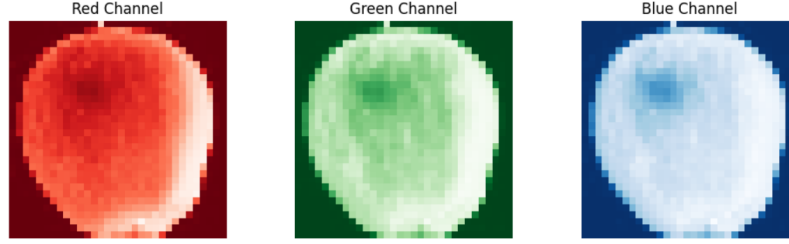
To effectively classify the fruit images, it is crucial to extract discriminative and informative features. We explored two prominent feature extraction techniques that capture complementary aspects of the images: color information and structural patterns.

2.1.1 Color Histogram

Color plays a significant role in distinguishing between different fruit categories. To leverage this, we utilized color histograms that represent the distribution of pixel intensities across the three color channels: Red, Green, and Blue (RGB). The extraction process involved:



- Dividing the intensity range (0–255) of each channel into 8 bins.
- Computing normalized histograms for the Red, Green, and Blue channels.



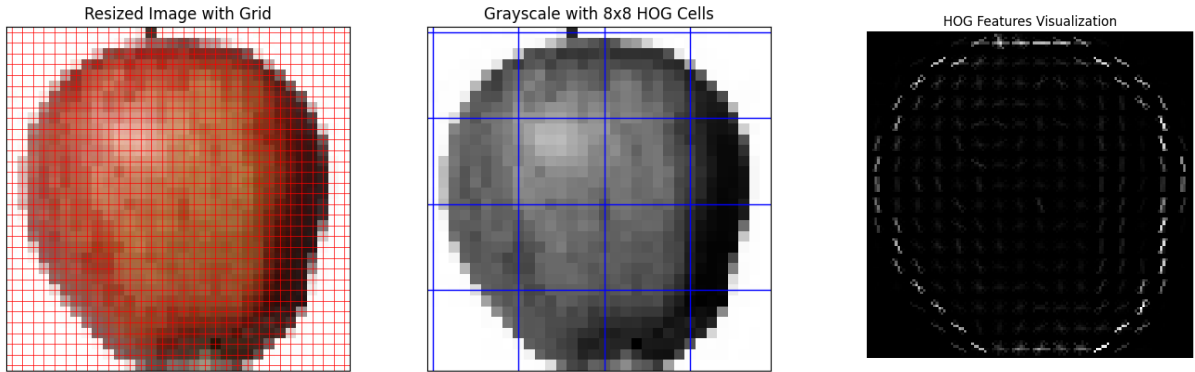
- Concatenating the histograms of all three channels into a single feature vector.

This method effectively captures the dominant color composition of each fruit, making it suitable for distinguishing fruits with unique color profiles.

2.1.2 Histogram of Oriented Gradients (HOG)

While color captures surface characteristics, shape and texture provide deeper insight into fruit structure. For this, we employed Histogram of Oriented Gradients (HOG), a feature descriptor that captures edge and contour information. The procedure involved:

- Resizing images to a fixed resolution of 32×32 pixels for consistency.
- Converting images to grayscale to focus on intensity gradients.
- Dividing each image into 8×8 pixel cells.
- Computing gradient directions and forming histograms within each cell.
- Grouping cells into overlapping blocks for normalization.
- Flattening the resulting descriptors into a feature vector.



The resulting HOG features effectively describe the structure and texture of the fruits, which helps to calculate the classification accurately.

2.2 Machine Learning Techniques Used

In this project, we used five different machine learning algorithms to classify fruit images based on extracted features. These techniques were evaluated to determine their effectiveness in terms of accuracy and generalization.

2.2.1 K-Nearest Neighbors (KNN) classifier

KNN -color histogram [Colab Notebook Link](#) KNN -HOG [Colab Notebook Link](#)

We implemented the K-Nearest Neighbors (KNN) algorithm to classify fruits based on their color histogram features. KNN is a simple yet effective classification technique that assigns a new sample to the most common class among its k-nearest training samples.

- **Model Initialization:** We implemented the K-Nearest Neighbors (KNN) classifier to evaluate both HOG and color histogram feature vectors extracted from fruit images.
- **Training the Model:** The model was trained using various values of k , where the training process involved storing the labeled feature vectors to be used for distance-based comparison during prediction.
- **Making Predictions:** For each test instance, the model calculated distances to all training instances, selected the k nearest neighbors, and assigned the most frequent class label among them.
- **Evaluating Performance:** We tested the model for different values of k and recorded the accuracy. The highest accuracy using HOG features was **85.34%** at $k = 1$, while the best performance for color histogram features was **94.94%** at $k = 1$. Accuracy declined as k increased, indicating sensitivity to the choice of k .

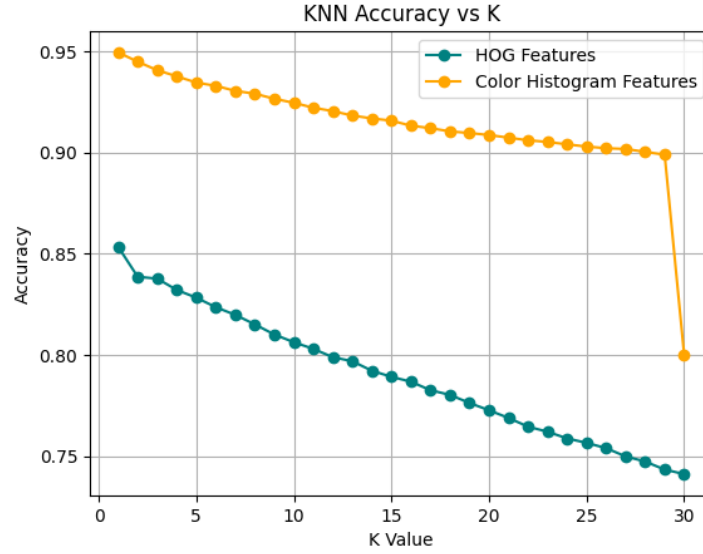


Figure 1: KNN Accuracy vs. k for HOG and Color Histogram

2.2.2 Decision Tree Classifier

Decision Tree Classifier -Color Histogram [Colab Notebook Link](#) Decision Tree Classifier -HOG [Colab Notebook Link](#) We implemented the Decision Tree classifier to classify fruits based on their extracted features. Decision Trees are widely used in classification tasks as they provide an interpretable model that makes decisions by recursively partitioning the feature space.

To train and evaluate the Decision Tree model, we followed these steps:

- **Data Preparation:** The fruit labels were encoded into numerical values using LabelEncoder, converting categorical class names into numbers.
- **Feature and Label Splitting:** The dataset was divided into features (X) and labels (y) for both training and testing sets.
- **Model Training:** A Decision Tree Classifier was initialized and trained using both the color histogram and HOG characteristics of the training set.
- **Making Predictions:** The trained model was then used to classify the fruits in the test dataset (X_{test}).
- **Performance Evaluation:** The predictions (y_{pred}) were compared with actual labels (y_{test}) to calculate the accuracy score.

- * **Color Histogram Features:** Achieved an accuracy of **86.75%**.
- * **HOG Features:** Achieved an accuracy of **61.00%**.

These results indicate that the Decision Tree performs significantly better with color histogram features compared to HOG, highlighting the importance of feature selection in classification tasks.

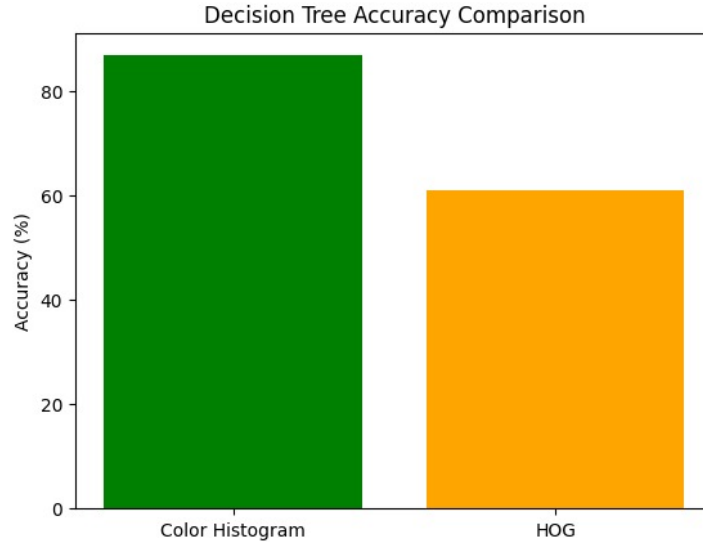


Figure 2: Accuracy comparison using different features with Decision Tree

2.2.3 Logistic Regression classifier

Logistic Regression Classifier -colour histogram [Colab Notebook Link](#) Logistic Regression classifier - HOG [colab Notebook Link](#)

We implemented the Logistic Regression classifier, a linear model used for binary and multi-class classification. This classifier models the probability of a sample belonging to a particular class using the logistic function.

The dataset was preprocessed by removing non-essential columns and encoding class labels into numerical values. It was then split into feature matrices. The process of training and evaluation follows these steps:

Model Initialization

HOG Features: A LogisticRegression classifier was initialized with the following hyperparameters: `max_iter=2000` ensures sufficient iterations for convergence, `solver='lbfgs'` optimizes the model efficiently for multi-class problems, `class_weight='balanced'` automatically adjusts weights for imbalanced classes, and `C=0.1` adds L2 regularization to prevent overfitting.

Color Histogram Features: A LogisticRegression model was used with `max_iter=1000`, which allows enough iterations for model convergence. Additionally, a polynomial feature transformation was applied using `PolynomialFeatures(degree=d, interaction_only=True)` where $d \in \{1, 2, 3\}$. This allowed the model to consider higher-order interactions between the histogram bins.

Both models rely on scikit-learn's internal optimization, which is based on probabilistic decision boundaries, without requiring manual specification of any loss function.

Training the Model

Color Histogram Features: The training pipeline involved dropping non-feature columns like filenames and labels, encoding class labels into integers using `LabelEncoder`, standardizing features

with StandardScaler, and transforming features with PolynomialFeatures for degrees 1, 2, and 3. Performance metrics were visualized to compare how polynomial degree impacts generalization.

HOG Features: The process included loading HOG feature vectors from CSV files, encoding class labels using LabelEncoder, standardizing feature vectors with StandardScaler, and fitting a logistic regression model on the scaled features.

Making Predictions

Both models rely on scikit-learn's internal optimization routines, which are based on probabilistic decision boundaries. Predictions are made by computing probability scores and then assigning the class corresponding to the highest score.

Performance Evaluation

The performance is assessed by comparing the predicted labels ($\hat{y}$) with the actual labels (y_{test}). This provides a quantitative evaluation of the model accuracy.

The Logistic Regression classifier achieved remarkable performance:

- Using **Color Histogram**, it achieved an accuracy of 95.45%.
- Using **Histogram of Oriented Gradients (HOG)**, it achieved an accuracy of 79.17%.

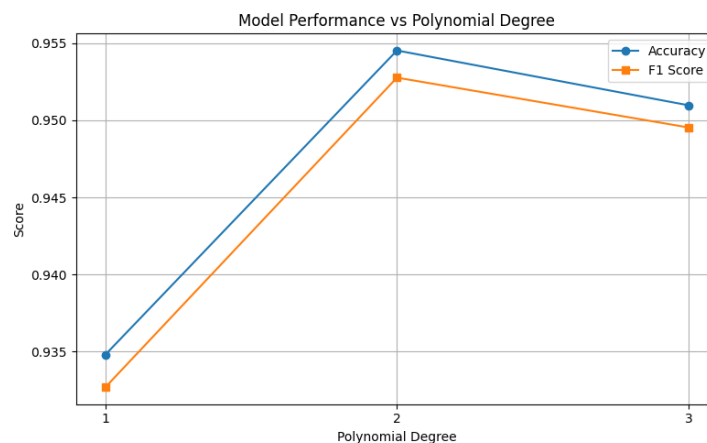


Figure 3: Accuracy comparison using Color Histogram features with Logistic Regression

2.2.4 Naïve Bayes Classifier

Naïve Bayes Classifier-Color Histogram Colab Notebook Link Naïve Bayes Classifier-HOG Colab Notebook Link

We implemented the Naïve Bayes classifier, a probabilistic algorithm based on Bayes' theorem. This classifier assumes that the features are conditionally independent given the class label, making it a simple yet effective method for classification tasks.

The dataset was preprocessed by removing non-essential columns and encoding class labels into numerical values. It was then split into feature matrices. The process of training and evaluation follows these steps:

- **Model Initialization:** We instantiated a Naïve Bayes classifier, specifically the Gaussian Naïve Bayes model, which assumes a normal distribution of feature values.
- **Training the Model:** The model was trained using the feature vectors and corresponding labels from the training set using color histogram.
- **Making Predictions:** Once trained, the model predicted class labels for the test samples based on posterior probabilities.

- **Evaluating Performance:** The classification accuracy was measured by comparing the predicted labels ($\hat{y}$) with the actual labels (y_{test}), providing a quantitative assessment of model performance.

The Naïve Bayes classifier achieved the following results:

- Using **Color Histogram**, it achieved an accuracy of 85.42%.
- Using **Histogram of Oriented Gradients (HOG)**, it achieved an accuracy of 55.27%.

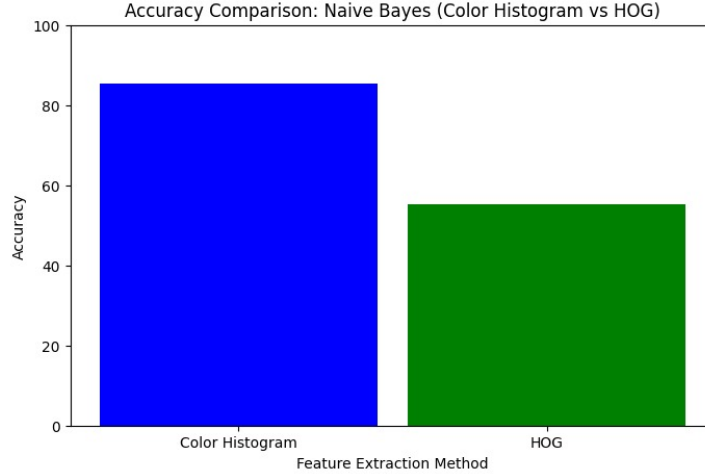


Figure 4: Accuracy comparison using different features with Naive Bayes

2.2.5 Support Vector Machine (SVM)

SVM-Color Histogram Colab Notebook Link SVM-HOG Colab Notebook Link

We implemented the **Support Vector Machine (SVM)** classifier, a supervised learning model used for classification tasks. SVM aims to find the optimal hyperplane that best separates the data into different classes. It is especially effective in high-dimensional spaces and performs well even when the number of dimensions exceeds the number of samples.

The dataset was preprocessed by removing irrelevant columns, encoding class labels numerically, and applying standardization to the features for better model performance. The workflow followed these steps:

- **Model Initialization:** We instantiated an SVM classifier with a linear kernel, suitable for linearly separable data and efficient for high-dimensional datasets.
- **Training the Model:** The SVM model was trained using standardized feature vectors and their corresponding class labels.
- **Making Predictions:** The trained model predicted class labels for the test samples based on the decision boundary learned during training.
- **Evaluating Performance:** The model's classification accuracy was measured by comparing predicted labels with actual test labels, offering a quantitative insight into the classifier's performance.

The SVM classifier achieved the following results:

- Using **Color Histogram** features, it achieved an **accuracy of 97.07%**.
- Using **Histogram of Oriented Gradients (HOG)** features, it achieved an **accuracy of 87.10%**.

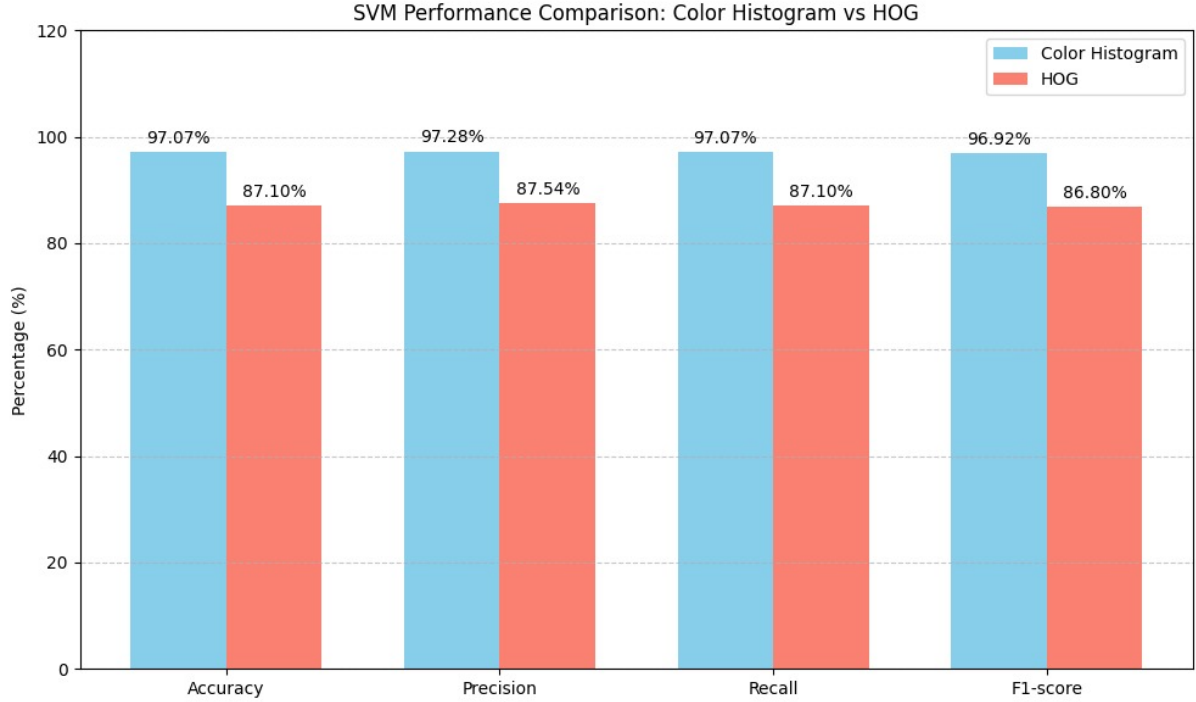


Figure 5: Accuracy comparison using different features with SVM

3 Experiments and Results

3.1 Dataset and Experimental Setup

The experiments were conducted using the Fruits 360 Dataset, which consists of labeled images of 141 fruit categories. The dataset was preprocessed by resizing images, normalizing pixel values, and extracting features using color histograms and Histogram of Oriented Gradients (HOG).

We implemented and tested five classification models: K-Nearest Neighbors (KNN), Decision Tree, Naïve Bayes, Support Vector Machine (SVM) and Logistic Regression using color histogram and HOG. Each model was trained using extracted features and evaluated based on accuracy. The experiments were conducted in Python using scikit-learn, and the models were optimized by tuning hyperparameters.

3.2 Comparison of Model Performance

To assess the effectiveness of each classifier, we computed accuracy scores on the test dataset. The results are summarized in Table 1.

Classifier	Accuracy-Color Histogram (%)	Accuracy-HOG(%)
K-Nearest Neighbors (KNN)	94.94	85.34
Decision Tree	86.75	61
Naïve Bayes	85.42	55.27
Logistic Regression	95.45	79.17
SVM	97.07	87.10

Table 1: Classification accuracy of different models on the test set.

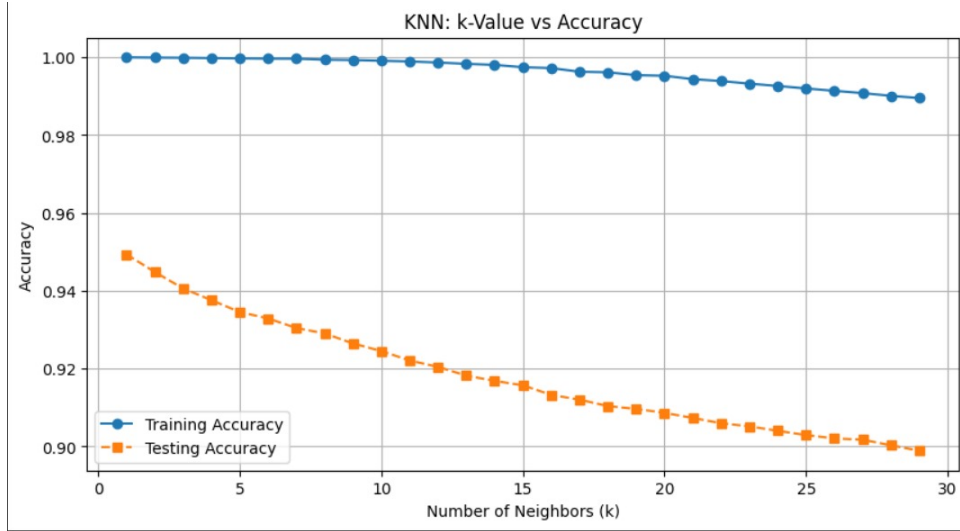


Figure 6: K-value vs. Accuracy for KNN

3.3 Analysis of Results

- **Support Vector Machine (SVM) (Color Histogram: 97.07%, HOG: 87.10%) – Best Performing Model:** SVM achieved the highest accuracy overall, especially excelling with Color Histogram features. Its strength in handling high-dimensional data and finding an optimal separating hyperplane made it the most effective classifier for this task. The slightly lower accuracy on HOG features may indicate that a different kernel (e.g., RBF) could further improve performance.
- **Logistic Regression (Color Histogram: 95.45%, HOG: 79.17%) – Strong Linear Model:** Logistic Regression also showed high performance, particularly with Color Histogram features. Its effectiveness in learning a linear decision boundary through gradient descent optimization contributed to consistent results. However, the drop in performance on HOG features suggests that these features might not align well with a purely linear model.
- **K-Nearest Neighbors (KNN) (Color Histogram: 94.94%, HOG: 85.34%) – Consistent and Competitive:** KNN showed strong accuracy with both types of features. Its instance-based approach is effective when the dataset has clear class separations. Minor performance dips with HOG may be due to higher dimensionality or sensitivity to noisy features.
- **Decision Tree (Color Histogram: 86.75%, HOG: 61%) – Prone to Overfitting:** Decision Tree performance was moderate with Color Histograms but significantly poorer with HOG features. This suggests the model may have overfitted to the training data and struggled to generalize, especially when handling complex or high-dimensional features.
- **Naïve Bayes (Color Histogram: 85.42%, HOG: 55.27%) – Struggled with Complex Features:** Naïve Bayes had the lowest accuracy across both feature types. Its assumption of feature independence likely hindered its effectiveness, especially with HOG features which often exhibit interdependencies. Despite its efficiency, it may not be suitable for datasets with complex feature interactions.

4 Summary

Support Vector Machine (SVM) emerged as the best-performing classifier, achieving the highest accuracy with both Color Histogram and HOG features. Logistic Regression and K-Nearest Neighbors (KNN) followed closely, showing consistent performance and indicating that the dataset exhibits structure suitable for both linear and distance-based classification. Decision Tree performance dropped significantly on HOG features, suggesting overfitting and poor generalization on high-dimensional features. Naïve Bayes had the lowest performance, likely due to its strong independence assumption not aligning with the feature distributions. Future work could explore

advanced ensemble methods, kernel tuning for SVM, and dimensionality reduction to further enhance classification performance.

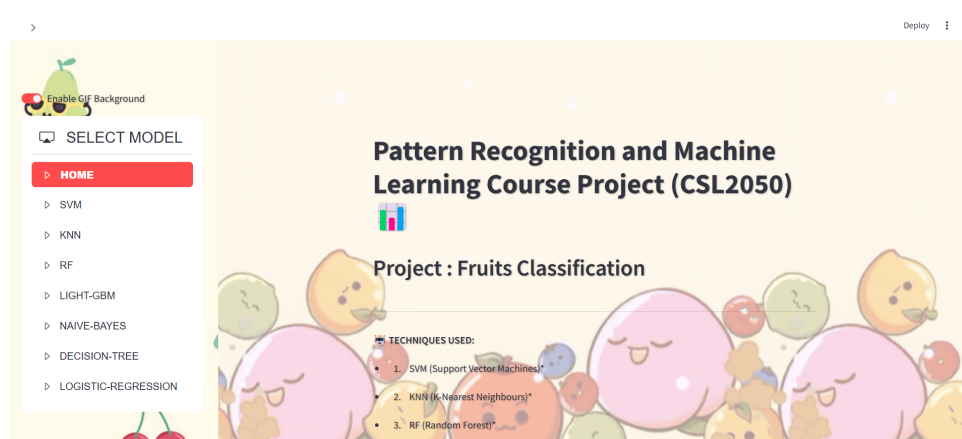


Figure 7: UI mainpage view

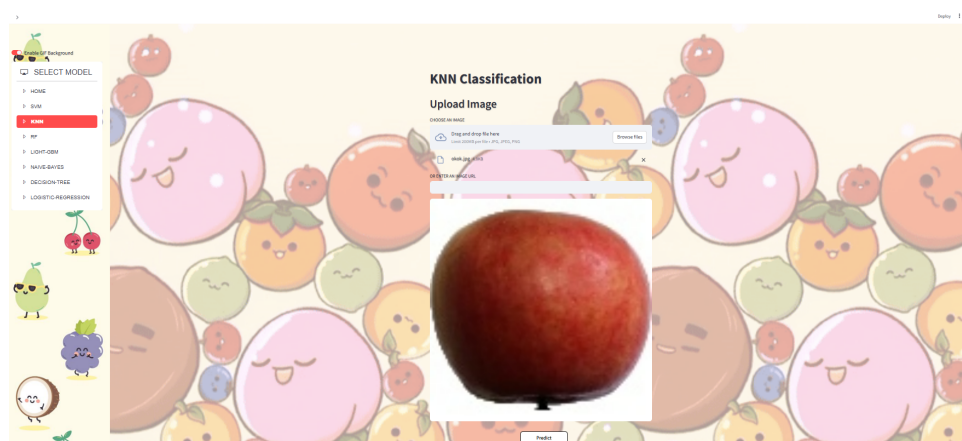


Figure 8: KNN UI view

This study references several important works [1, 2, 3, 4, 5, 6, 7].

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4 Contribution of Each Member

1. **B23CM1017**: Worked on Decision Tree Model using Colour Histogram features and HOG features , contributed to feature extraction from the Fruits 360 dataset, deployed the webpage.
2. **B23CM1024**: Worked on Logistic Regression using Colour Histogram features and HOG features , contributed to feature extraction from the Fruits 360 dataset, prepared power point presentation.
3. **B23CS1052**: Worked on KNN using Colour Histogram features and HOG features , designed the basic web page structure, Git hub Readme file and Minutes of meeting.
4. **B23EE1021**: Enhanced UI, completed the entire front-end section, worked on SVM model using Colour Histogram features and HOG features , Mid project report, created Youtube video.
5. **B23EE1077**: Worked on the Naive Bayes Model using Colour Histogram features and HOG features, contributed to feature extraction from the Fruits 360 dataset, Final project report. Each member contributed in making the report, ppt and youtube video recoding at their individual ML models mentioned above.